SWA-DPBR: Socially-Weighted Alignment with Dual-Pass Bootstrap Reliability for Cross-Cultural Moral Alignment of Large Language Models

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Abstract

How far can we align a frozen LLM to *different* cultures *without* training? We target **training-free cross-cultural alignment at the logit level**: no weight updates, no per-country finetuning—only a scalar decision-gap correction computed at inference time so that binary moral choices track country-specific human preference vectors instead of a single global default. **SWA-DPBR** (Socially-Weighted Alignment with Dual-Pass Bootstrap Reliability) realizes this by letting WVS-grounded personas disagree within each country, converting that disagreement into a bounded logit shift via Prospect-Theory importance sampling, merging two independent IS passes with a bootstrap reliability weight, and applying a light hierarchical prior—so unstable sampling automatically shrinks aggressive updates. On MultiTP, across twenty countries on four continents and open models spanning several families (about 3.8B–70B parameters), strong checkpoints cut aggregate misalignment versus vanilla by roughly 19–24%, with gains spread across the Americas, Europe, and Asia rather than isolating to one geography. When steering fails, the pattern is ill-conditioned *decision* logits under importance weighting—an architectural bottleneck, not a blind spot toward entire world regions—which suggests that logit-space, inference-time cultural steering is a practical lever whenever the base model's decision surface is well behaved. Code: `[repository]`.

1 Introduction

When a self-driving car must choose between sparing a group of elderly men or a group of young women, there is no culturally universal answer. The Moral Machine experiment [Awad et al., 2018] collected 40 million judgments from 233 countries and revealed that such preferences vary profoundly across cultures: East Asian respondents weight youth more heavily; Southern-hemisphere populations show distinct utilitarian calculi; Islamic societies prioritise different social roles. Yet the large language models increasingly enlisted to assist with such decisions encode a single, culturally monolithic moral stance inherited from their predominantly Western training data [Henrich et al., 2010].

The multilingual successor benchmark MultiTP [Jin et al., 2025] operationalises this evidence into country-specific Average Marginal Component Effects (AMCE) across six moral dimensions and 107 languages, providing a rigorous test bed for cross-cultural alignment. The verdict is stark: Jin et al. [2025] found that none of 19 contemporary LLMs achieved broadly satisfactory alignment; most exhibit WEIRD-biased reasoning that departs markedly from non-WEIRD human preferences.

How should this gap be closed? A useful cross-cultural decoder would ideally satisfy three constraints at once: *no* weight updates, *no* per-culture supervised labels at scale, and *no* white-box access to

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internal activations. Existing work typically trades one of these away. *Fine-tuning* [Bai et al., 2022, Ouyang et al., 2022] changes weights and needs per-culture data. *Prompting-based steering* avoids training but lacks empirical grounding [Jin et al., 2025]. *Activation steering* [Arditi et al., 2025, Zou et al., 2023] needs white-box access and bespoke engineering. *Reward-guided decoding* [Khanov et al., 2024, Mudgal et al., 2024] assumes reward models we do not have for every country. *Culture-aware adapters* [Li et al., 2024, Yao et al., 2024] require per-culture training and scale poorly across 100+ countries.

We propose **SWA-DPBR** (Socially-Weighted Alignment with Dual-Pass Bootstrap Reliability) to meet all three constraints. The central idea is that *disagreement is signal*. Moral preferences within a society are not monolithic-age cohorts and strata prioritise different trade-offs [Awad et al., 2018, Kleiman-Weiner et al., 2017]. The World Values Survey [Haerpfer et al., 2022] supplies empirical structure for that within-country variation. We instantiate each country as a small panel of culturally grounded persona agents; when those agents *disagree* on a dilemma, the spread of their logit gaps *localises* where a correction is plausibly useful, and when they agree, the vanilla model is often already close to consensus. When they disagree strongly, a single aggregate "answer" is the wrong target.

Operationally, SWA-DPBR builds a fixed WVS-grounded persona ensemble per country, evaluates base plus personas in one batched forward pass with native-language prompting, applies positional debiasing, and runs two independent Prospect-Theory-weighted importance-sampling passes [Kahneman and Tversky, 1979] over consensus perturbations. If the two passes yield similar corrections, we merge them; if they diverge, a bootstrap-style reliability weight shrinks the update, and an effective-sample-size guard can anchor unstable estimates back toward the debiased base logit before a light hierarchical country prior. Full equations are in §3; default hyperparameter values are in Table 7.

Contributions. Our contributions are threefold:

- **Training-free, inference-time cultural alignment.** We introduce SWA-DPBR, which aligns a *frozen* LLM to country-level human preference vectors without weight updates: WVS-grounded personas encode within-country variation; positional debiasing cancels framing bias; two independent Prospect-Theory importance-sampling passes estimate a consensus perturbation; a bootstrap reliability weight merges them; and a light hierarchical prior smooths corrections across scenarios (§3).
- **Multi-country evaluation across scales and model families.** On MultiTP we study twenty countries and open checkpoints from multiple families from about 3.8B to 70B parameters, reporting large macro gains versus vanilla, per-country geographic structure, and robustness checks that support the headline numbers (§5).
- **Ablations and robustness.** Controlled slices show that persona diversity and debiasing are load-bearing, that dropping to consensus-only or simpler utility kernels hurts, and that temperature and preprocessing sweeps leave headline conclusions stable (§6; Appendices E–F).

2 Related Work

Cross-cultural moral benchmarks. The Moral Machine experiment [Awad et al., 2018] provides 40M+ cross-cultural moral judgments. MultiTP [Jin et al., 2025] extends this to 107 languages with country-specific AMCEs; we adopt it as our benchmark and propose an alignment *intervention* rather than diagnostics alone. Other benchmarks (ETHICS [Hendrycks et al., 2021], MoralExceptQA [Jin et al., 2022], MoCa [Nie et al., 2023]) remain English-centric. PRISM [Kirk et al., 2024] and GlobalOpinionQA [Durmus et al., 2023] address diversity but cover few languages. Recent frameworks (EvalMORAAL [Aakanksha et al., 2024], CARB [Myung et al., 2024]) emphasise broader-topic evaluations; SWA-DPBR targets binary moral decisions specifically. CROSS [Qiu et al., 2025] evaluates culturally grounded multimodal safety; we remain text-only with trolley vignettes so country-level AMCE vectors stay directly comparable to the model's six-dimensional output.

Cross-cultural LLM evaluation. LLM outputs correlate most strongly with liberal, educated American respondents [Santurkar et al., 2023]. RLHF introduces unintended cultural impacts [Ryan

et al., 2024], and assigning cultural personas can surface toxic outputs [Deshpande et al., 2023]. This recurring WEIRD bias motivates SWA-DPBR's calibration objective.

Inference-time alignment. Activation steering [Arditi et al., 2025, Zou et al., 2023] requires white-box access. ARGS [Khanov et al., 2024] and controlled decoding [Mudgal et al., 2024] need trained reward models unavailable for cross-cultural settings (Appendix H). CultureManager [Li et al., 2024] and CAReDiO [Yao et al., 2024] train culture-specific adapters but require per-culture labelled data and weight modification, scaling poorly across 100+ countries. To the best of our knowledge, SWA-DPBR is the first training-free method targeting country-level moral calibration specifically. Sorensen et al. [2024] articulate the goal of pluralistic alignment, which SWA-DPBR operationalises concretely. Related exploratory runs in our codebase suggest that *contrastive* persona formulations—explicitly amplifying a country direction against a world anchor—can shatter alignment on already miscalibrated backbones; SWA-DPBR therefore stays with direct WVS personas and logit-space PT–IS rather than contrastive amplification. Recent work shows that inference-time stochasticity can improve moral alignment scores on trolley-style tasks by reshaping confidence [Kwon et al., 2025]; SWA-DPBR is complementary in targeting *directional*, survey-grounded, country-specific corrections at logits rather than generic uncertainty inflation. A controlled head-to-head on the same MultiTP grid is important future work.

Positioning against adjacent inference-time methods. SWA-DPBR targets a narrow slice: *black-box*, single-model, binary forced-choice moral decisions with country-conditioned survey personas-no extra trained aligner, unlike *DIFFPO*-style policy-agnostic aligners; no fixed global steering vector, unlike *SWAI*-style attribute shifts; no internal-layer access, unlike *SLED*-style white-box methods. Structured multi-agent systems such as OG-MAR [Zhang et al., 2024] pursue richer ontology-level reasoning; we trade that depth for one additional batched forward pass and explicit country-level calibration.

Importance sampling, reliability, and Prospect Theory. Self-normalised importance sampling is biased at finite K; splitting the same Monte Carlo budget into two independent replicates and comparing their outputs is a standard way to expose sampling noise without doubling wall-clock forward passes at the transformer level (each replicate reuses cached persona logits). We combine that diagnostic with a Prospect-Theory utility [Kahneman and Tversky, 1979] on logit-gap coordinates, following the control-as-inference view used in path-integral and KL-regularised steering work [Williams et al., 2018, Huang et al., 2024]. Dual-pass reliability is a variance-based heuristic, not an unbiasedness certificate; we report it as part of the method's explicit uncertainty model.

3 Method: SWA-DPBR

3.1 Problem Formulation

Let $\mathcal{X}$ be the set of moral dilemma scenarios, each presenting a forced binary choice via decision tokens `A` (a) and `B` (b). For target country c, let $\mathbf{h}^{(c)} \in \mathbb{R}^6$ be the human AMCE vector from 40M+ judgments [Awad et al., 2018]. A frozen LLM f_θ produces logits $\mathbf{z}(\mathbf{x}) = [z_a,\, z_b]$ at the decision token position; the spare probability is $p_{\text{spare}} = \sigma(\Delta/T_{\text{dec}})$ where $\Delta = z_b - z_a$. Each dimension's AMCE is estimated as:

$$\hat{m}_d^{(c)} = \frac{1}{|\mathcal{S}_d^{(c)}|} \sum_{\mathbf{x} \in \mathcal{S}_d^{(c)}} p_{\text{spare}}(\mathbf{x}), \tag{1}$$

following the MultiTP evaluation convention [Jin et al., 2025]. For the five binary dimensions (Species, Gender, Age, Fitness, Social Value), this reduces to the empirical mean of p_{spare}. For Utilitarianism the predictor is the count difference $n_{\text{diff}} = n_{\text{pref}} - n_{\text{non-pref}}$; we fit $p_{\text{spare}}(\mathbf{x}) = a + b\, n_{\text{diff}}(\mathbf{x}) + \varepsilon$ by OLS and report $\hat{m}_{\text{Util}}^{(c)} = a + b\, \overline{n_{\text{diff}}}$, i.e. the fitted value at the mean observed count difference (Appendix A).

Goal: Without modifying θ, find a logit correction $\delta^*(\mathbf{x}, c)$ such that the corrected AMCE vector $\hat{\mathbf{m}}^{(c)}(\theta, \delta^*)$ is closer to $\mathbf{h}^{(c)}$. We primarily measure alignment with the *misalignment score* (MIS): ℓ_2 distance between $\hat{\mathbf{m}}^{(c)}$ and $\mathbf{h}^{(c)}$ (lower is better). Jensen–Shannon divergence and Pearson r are reported as secondary diagnostics where noted.

3.2 Cultural Persona Construction

For each country c, SWA-DPBR constructs $N = 4$ persona agents from WVS Wave 7 individual-level microdata [Haerpfer et al., 2022]. We extract 8 value dimensions from specific WVS items: *gender egalitarianism* (Q58P, Q59P, Q60P), *religious importance* (Q6P), *interpersonal trust* (Q43P), *moral permissiveness* (Q50, Q52P, Q54P), *work centrality* (Q5P), *family importance* (Q1P), *personal autonomy* (Q39P), and *meritocratic orientation* (Q40P). For each country, respondents are segmented into three age cohorts using birth year and survey year: *young* (18–35), *middle-aged* (36–55), and *older* (>55). Dimension means are computed per cohort and converted into natural-language descriptors via fixed thresholds (Appendix B). A fourth, country-invariant *utilitarian agent* maximises lives saved. Each persona is a natural-language system prompt:

> *You are a [age-cohort] adult from [country]. The cultural values typical of your social group are: [WVS statistics]. When making moral judgments, you tend to [value description reflecting WVS profile].*

WVS grounding ensures empirically measured within-country variation rather than stereotypes [Deshpande et al., 2023]. Concrete persona examples and the WVS-to-trolley dimension linkage are detailed in Appendix B.

3.3 Batched Evaluation and Logit Extraction

All $N + 1 = 5$ input sequences (base + 4 personas) are evaluated in a *single* batched forward pass:

$$\mathbf{z}_{\text{base}},\ \mathbf{z}_1, \ldots, \mathbf{z}_N = f_\theta(\mathbf{x};\ \varnothing,\ \mathbf{s}_1, \ldots, \mathbf{s}_N)\,. \tag{2}$$

Per-category logit temperatures T_{cat} scale logits before computing decision gaps. We use $T_{\text{cat}} = 4.0$ (Species), 3.5 (Gender), and 1.5 for Age, Fitness, Social Value, and Utilitarianism (Table 1); the same values appear in Appendix D.

$$\delta_i = \frac{z_{i,b} - z_{i,a}}{T_{\text{cat}}}, \quad \delta_{\text{base}} = \frac{z_{\text{base},b} - z_{\text{base},a}}{T_{\text{cat}}}, \quad \bar{\delta} = \frac{1}{N}\sum_{i=1}^{N} \delta_i \quad \text{(consensus)}, \tag{3}$$

$$r_i = \delta_i - \delta_{\text{base}} \quad \text{(per-agent reward)}. \tag{4}$$

Here δ_{base} is the decision gap of the base prompt (no persona). The reward r_i measures how strongly persona i shifts the model *away from the culturally-neutral baseline*: a large $|r_i|$ indicates that the persona has identified a direction of moral preference the vanilla model does not exhibit. Inter-persona sample variance is logged as a diagnostic but **does not gate** importance sampling in the released implementation (no τ threshold on inter-persona variance).

3.4 Positional Debiasing

To cancel recency bias [Liu et al., 2024], each scenario is evaluated twice: once in the original ordering, once with *both* choice labels (A ↔ B) and group labels (Group A ↔ Group B) swapped in the native language. The debiased decision gap for each agent is $\delta_i = (\delta_i^{(\text{orig})} - \delta_i^{(\text{swap})})/2$; the base prompt yields δ_{base} under the same rule.

3.5 Prospect-Theory IS with Dual-Pass Reliability

SWA-DPBR runs two *independent* importance-sampling passes on every scenario. Each pass draws $K_{\text{half}} = K/2 = 64$ perturbations $\epsilon_k \sim \mathcal{N}(0, \sigma^2)$; σ defaults to 0.3 but is lifted to at least the empirical standard deviation of persona gaps when that is larger. Candidates are $\tilde{\delta}_k = \bar{\delta} + \epsilon_k$.

Per-persona gains and collective utility. For candidate $\tilde{\delta}_k = \bar{\delta} + \epsilon_k$, define *scalar gains* in logit-gap units that measure how much moving the consensus toward $\tilde{\delta}_k$ reduces L^1 distance to each persona relative to the base model:

$$g_{i,k} = |\delta_{\text{base}} - \delta_i| - |\tilde{\delta}_k - \delta_i|, \tag{5}$$

$$g_{\text{cons},k} = |\delta_{\text{base}} - \bar{\delta}| - |\tilde{\delta}_k - \bar{\delta}|. \tag{6}$$

Large $g_{i,k}$ means $\tilde{\delta}_k$ is closer than δ_{base} to persona i's debiased gap; $g_{\text{cons},k}$ is the same idea with the persona consensus $\bar{\delta}$ as the reference. Prospect Theory is applied *per persona before averaging*, preserving loss aversion at the social level [Kahneman and Tversky, 1979]:

$$U_{\text{total}}(\epsilon_k) = (1 - \lambda_{\text{coop}}) \frac{1}{N} \sum_{i=1}^{N} v\Big(\frac{g_{i,k}}{\sigma}\Big) \; + \; \lambda_{\text{coop}}\, v\Big(\frac{g_{\text{cons},k}}{\sigma}\Big), \tag{7}$$

with $\lambda_{\text{coop}} = 0.7$. Dividing by σ makes the PT input dimensionless and comparable across moral dimensions (as in the released controller). The per-agent offsets $r_i = \delta_i - \delta_{\text{base}}$ from Eq. 4 remain useful diagnostics but **do not** enter U_{total} in this implementation.

$$v(x) = \begin{cases} x^{\alpha} & x \geq 0, \\ -\kappa\, |x|^{\beta} & x < 0. \end{cases} \tag{8}$$

We use $\alpha = \beta = 0.88$ and $\kappa = 2.25$ [Kahneman and Tversky, 1979]. There is **no** separate quadratic penalty on ϵ_k in the released utility: the Gaussian proposal $\mathcal{N}(0, \sigma^2)$ already limits step size, and an explicit KL-on-ϵ term is omitted (see Appendix C).

Per-pass IS update and ESS guard. For each pass $m \in \{1, 2\}$ we form softmax weights over its own K_{half} samples,

$$w_k^{(m)} = \frac{\exp\big(U_{\text{total}}(\epsilon_k)/\eta\big)}{\sum_{k'} \exp\big(U_{\text{total}}(\epsilon_{k'})/\eta\big)}, \qquad \delta_m^{\star} = \sum_{k=1}^{K_{\text{half}}} w_k^{(m)}\, \epsilon_k, \tag{9}$$

with softmax inverse temperature $\eta = 0.5$ (field `temperature` / `mppi_temperature` in `SWAConfig`; distinct from T_{dec} scaling the final sigmoid, though we also set $T_{\text{dec}} = 0.5$). Let $N_{\text{eff}}^{(m)} = 1/\sum_{k=1}^{K_{\text{half}}} (w_k^{(m)})^2$ (Kish effective sample size for normalised weights) and define the **normalised** ratio $\widehat{\text{ess}}^{(m)} = N_{\text{eff}}^{(m)}/K_{\text{half}} = 1/(K_{\text{half}} \sum_k (w_k^{(m)})^2)$. If $\widehat{\text{ess}}^{(m)} < \rho_{\text{eff}}$ we set $\delta_m^{\star} \leftarrow 0$ (binary guard; default $\rho_{\text{eff}} = 0.1$).

Bootstrap reliability and anchor blend. Define bootstrap variance $v_{\text{b}} = (\delta_1^{\star} - \delta_2^{\star})^2$ and soft reliability $r = \exp(-v_{\text{b}}/s)$ with scale s=0.04 (`VAR_SCALE`). The merged IS offset is

$$\delta_{\text{IS}}^{\star} \; = \; r \cdot \frac{\delta_1^{\star} + \delta_2^{\star}}{2}. \tag{10}$$

Large disagreement between passes shrinks r and pulls the correction toward zero without a hard gate. Finally, with $\text{ess}_{\min} = \min(\widehat{\text{ess}}^{(1)}, \widehat{\text{ess}}^{(2)})$ and lower clip ρ_{eff}, set $\alpha_{\text{reg}} = \text{clip}(\text{ess}_{\min},\, \rho_{\text{eff}},\, 1)$ and form the micro correction before the hierarchical prior,

$$\delta_{\text{micro}} = \alpha_{\text{reg}}\, \bar{\delta} + (1 - \alpha_{\text{reg}})\, \delta_{\text{base}} + \delta_{\text{IS}}^{\star}, \tag{11}$$

where δ_{base} is the symmetric swap average for the base prompt (§3.4). When the ESS-anchor blend is *enabled* (default in the released code), Eq. 11 applies as written. When it is *disabled*, the implementation sets $\delta_{\text{micro}} = \bar{\delta} + \delta_{\text{IS}}^{\star}$ only (no interpolation toward δ_{base}). Geometrically, $\alpha_{\text{reg}} = \text{clip}(\text{ess}_{\min},\, \rho_{\text{eff}},\, 1)$ interpolates between the persona consensus $\bar{\delta}$ and the debiased base δ_{base}: when both IS passes are diffuse ($\widehat{\text{ess}}^{(m)} \approx 1$), $\alpha_{\text{reg}} \approx 1$ and the consensus term dominates; when either pass collapses, α_{reg} approaches ρ_{eff} and the update leans toward δ_{base}. All MultiTP numbers in this paper use the *enabled* path unless noted.

3.6 Hierarchical country prior

After δ_{micro}, a scalar *country-level* smoother runs per ISO-3 key. Let $t \in \{0, 1, \ldots\}$ index scenarios in processing order for country c, and maintain an EMA statistic δ_{country}:

$$\delta_{\text{country}} \leftarrow (1 - \beta_{\text{ema}})\, \delta_{\text{country}} + \beta_{\text{ema}}\, \delta_{\text{micro}}, \tag{12}$$

with $\beta_{\text{ema}} = 0.1$, updated after each scenario. Let $\alpha_h(t)$ be an annealed blend weight,

$$\alpha_h(t) = \begin{cases} 0, & t < N_{\text{warm}}, \\ 1 - \exp\big(-(t - N_{\text{warm}})/\tau\big), & t \geq N_{\text{warm}}, \end{cases} \tag{13}$$

Algorithm 1 SWA-DPBR Inference for Scenario $\mathbf{x}$ in Country c

Require: Frozen LLM f_θ; personas $\{\mathbf{s}_i\}_{i=1}^N$; hyperparameters $K_{\text{half}}, \sigma, \lambda_{\text{coop}}, \eta, T_{\text{dec}}, s$
Ensure: $p_{\text{spare}} \in [0, 1]$

```
for order ∈ {orig, swap} do
    Batched forward: evaluate f_θ with base + N personas
    Extract logit pairs; compute δ_i^(ord)                          ▷ Eq. 2
end for
Average: δ_i ← (δ_i^(orig) − δ_i^(swap))/2                          ▷ §3.4
Compute consensus δ̄, rewards r_i                                   ▷ Eqs. 3–4
for m ∈ {1, 2} do
    for k = 1 to K_half do
        Sample ε_k ~ N(0, σ²); compute U_total(ε_k)                 ▷ Eqs. 5–7
    end for
    δ*_m ← Σ_k w_k^(m) ε_k or 0 if ess̃^(m) < ρ_eff                  ▷ Eq. 9
end for
r ← exp(−(δ*_1 − δ*_2)²/s); δ*_IS ← r(δ*_1 + δ*_2)/2                ▷ Eq. 10
δ_micro ← blend per Eq. 11 or δ̄ + δ*_IS if ESS-anchor off           ▷ see §3.5
δ_final ← country prior via Eqs. 12–14
return p_spare = σ(δ_final/T_dec)
```

SWA-DPBR pipeline overview (schematic).

Left: WVS Persona Construction → Middle: Batched Forward Pass → Right: Dual-pass PT-IS + reliability merge

Figure 1: **SWA-DPBR pipeline. Left:** WVS-grounded persona construction (3 age cohorts + utilitarian). **Middle:** Batched forward pass with positional debiasing. **Right:** Two independent IS passes merged with bootstrap reliability r (Eq. 10).

with $N_{\text{warm}} = 50$ and $\tau = 100$. Then

$$\delta_{\text{final}} = \alpha_h(t)\, \delta_{\text{country}} + \big(1 - \alpha_h(t)\big)\, \delta_{\text{micro}}. \tag{14}$$

For $t < N_{\text{warm}}$ the prior is inactive ($\delta_{\text{final}} = \delta_{\text{micro}}$); thereafter the blend gradually shifts mass onto the EMA of past micro corrections. Isolation ablations that switch off α_h or freeze δ_{country} are not tabulated here.

The final spare probability is $p_{\text{spare}} = \sigma(\delta_{\text{final}}/T_{\text{dec}})$ with $T_{\text{dec}} = 0.5$.

Algorithm 1 summarises the full procedure; numerical values for all symbols appear in Table 7 (Appendix D).

Table 1: The six moral dimensions in MultiTP.

Dimension	Preferred	Contra	T_{cat}
Species	Human	Animal	4.0
Gender	Female	Male	3.5
Age	Young	Elderly	1.5
Fitness	Fit	Unfit	1.5
Social Value	High (exec.)	Low (homeless)	1.5
Utilitarianism	More lives	Fewer lives	1.5

4 Experimental Setup

4.1 Dataset

We evaluate on MultiTP [Jin et al., 2025]: trolley-problem vignettes in native languages across six moral dimensions (Table 1). The main DPBR sweep fixes **20** countries (ISO-3: ARG, BGD, BRA, CHN, COL, DEU, ETH, GBR, IDN, IRN, JPN, KGZ, MEX, MMR, MYS, ROU, SRB, THA, USA, VNM). For each country we apply: (i) deduplication (retain only original-phrasing vignettes), (ii) quality filtering (remove Utilitarianism scenarios where both groups have equal size, as these carry no utilitarian signal), (iii) category capping (max 80 per dimension), and (iv) up-sampling via random oversampling with replacement (min 36 per dimension for stable AMCE estimation), yielding 310–500 scenarios per country (Appendix F).

4.2 Models

The main results focus on six open-weight checkpoints with robust macro gains on MultiTP: **Llama-3.3-70B**, **Magistral-Small-2509** (24B), **Phi-4** (14B), **Qwen3-VL-8B**, **Qwen2.5-7B**, and **Phi-3.5-mini** (3.8B). Table 2 orders them by *nominal* parameter count (descending); we evaluated additional checkpoints (smaller, quantised, or alternate inference setups) during development but omit those rows here when macro improvement was negligible or unstable under importance sampling. Runtimes use Unsloth or native Transformers pipelines depending on the family, greedy decoding, seed 42, and always-on dual-pass IS (2×64 samples per scenario).

Native-language evaluation. Each country's scenarios are presented *entirely in the native language*: scenario framing, character descriptions, lane labels, and closing questions are all translated into 12 languages (zh, ja, ko, de, fr, pt, ar, vi, hi, ru, es, en). Only the *decision tokens* (`A`/`B`) remain English; these are verified as single-token entries in each model's vocabulary (Appendix G). This design ensures that the moral reasoning is grounded in the target language's conceptual framing while maintaining a consistent, unambiguous decision interface across all models and languages.

Engineering lineage. The released configuration is the endpoint of iterative engineering over the same hierarchical persona–importance-sampling stack: variants explored adaptive anchor blending, entropy-based σ, Nesterov-style IS, category-coherence regularisers, per-dimension priors, and related knobs. On a fixed three-backbone, five-country prototyping panel (Qwen2.5-7B, Gemma-2-9B, Mistral-7B), dual-pass bootstrap reliability ranked among the strongest candidates on mean MIS while materially lowering IS flip rates versus earlier single-pass baselines—we prioritised that stability–accuracy tradeoff before scaling to the twenty-country paper sweep.

4.3 Baselines and metrics

On Llama-3.1-70B we compare (1) **Vanilla** (generic system prompt), (2) **Country-Tailored Prompt (B1)**, (3) **WVS Profile Prompting (B2)**, (4) **PRISM-Style Prompting (B3)** [Kirk et al., 2024], (5) **Activation Steering** [Arditi et al., 2025] with culturally contrastive steering vectors (implementation: Appendix H). Reward-guided methods (ARGS [Khanov et al., 2024], controlled decoding [Mudgal et al., 2024]) are excluded because they require reward models we do not have for cross-cultural targets (Appendix H). We likewise omit a full twenty-country run of inference-time dropout calibration [Kwon et al., 2025] and of pure temperature/margin scaling as standalone baselines; Appendix E reports structured sensitivity of T_{dec} and T_{cat} that partially overlaps simpler scaling ablations, and we flag direct comparators as future work. Primary headline metric is **MIS** (misalignment score):

Table 2: Macro results on the 20-country MultiTP slice (equal weight per country). MIS is mean ℓ_2 AMCE distance (↓). Rows follow nominal parameter count (descending); gains vs. vanilla are for reference only. "Win" counts countries with lower DPBR MIS than vanilla; mean r averages per-country Pearson correlation between model and human AMCE vectors.

Model	Mean MIS ↓	vs. van. (%)	Win / 20	Mean r
Llama-3.3-70B	0.668	+21.4	**20**	-0.01
Magistral-Small-2509 (24B)	0.350	+13.1	16	$+0.62$
Phi-4 (14B)	**0.346**	+23.7	18	$+0.56$
Qwen3-VL-8B	0.466	+18.9	**20**	$+0.22$
Qwen2.5-7B	0.362	+20.1	17	$+0.45$
Phi-3.5-mini (3.8B)	0.571	+10.4	17	-0.35

the ℓ_2 distance between the model's six-dimensional AMCE vector and the human vector for the same country [Jin et al., 2025], computed as in §3.1. Lower is better. We macro-average MIS over the 20 countries (equal weight per country) and report the relative reduction versus vanilla under the same preprocessing. **Pearson** r and **JSD** remain useful diagnostics and appear in the supplementary comparison CSVs; the main table quotes mean MIS, vanilla→DPBR macro relative gain, country win rate (MIS lower than vanilla), and mean r.

5 Results: Strong Gains Across Models and Countries

5.1 Cross-Model Comparison (20 Countries)

The central empirical question is whether *training-free* logit steering can materially close the gap to human AMCE vectors when we hold everything else fixed and only change the decoding pipeline. Table 2 answers that question affirmatively for six open checkpoints we highlight after a wider exploratory sweep. Rows are sorted by nominal parameter count rather than by gain, so the reader sees families side by side: the right-hand columns then show that SWA-DPBR improves macro MIS versus vanilla by substantial relative margins for every model in the panel, with country-level wins on most rows for most checkpoints—evidence that the effect is not a one-off quirk of a single backbone or a single geography.

Two patterns make this more than a leaderboard story. First, *calibration competes with scale*: the lowest absolute MIS in the table comes from a mid-sized instruction model whose logits align well with the IS stage, while a 70B-class model achieves a full sweep of per-country wins despite starting from a weaker vanilla profile—showing that DPBR can *rescue* a misaligned base as well as *refine* an already strong one. Second, the smallest checkpoint we report (Phi-3.5-mini) still posts a clear aggregate gain, which supports the practical claim that inference-time cultural steering is not reserved for frontier-scale models. Additional model–country combinations beyond the main panel include stacks where gains are smaller or mixed; we defer that frontier to §7 and Appendix K, because the headline takeaway here is that the method's *typical* behaviour on well-behaved logits is large, consistent, and portable across families.

Reference. For context, an earlier single-configuration prototype in our lab reached mean MIS near 0.4 on a comparable slice; the six rows here mix scales and therefore sit higher on average—illustrating variance across families rather than a single universal ceiling.

5.2 Geographic Structure and Per-Country Grids

Table 3 shows the same six checkpoints at the level of individual countries, grouped by region so the macro gains translate into a *map-level* story. The consistent takeaway is breadth: for strong backbones, improvements are visible across the Americas and Europe, track strongly through Southeast Asia when the base model already carries usable signal, and remain heterogeneous across East Asia in exactly the way one would expect when the same pipeline is applied to models with different off-manifold behaviour. In other words, SWA-DPBR does not "only work in the West"—it tracks where the human targets and the model logits are simultaneously informative.

The singleton panels (West Asia and Africa) are the natural stress tests: WVS persona coverage is sparser, so variance is higher and a small backbone can occasionally move the wrong direction on a

Table 3: **Per-country SWA-DPBR results (20 countries).** Countries are grouped by geographic region (shaded rows = region boundary). Vanilla and SWA-DPBR MIS (↓), relative MIS improvement (%), and SWA-DPBR JSD (↓) / Pearson r (↑). Models in the same order as Table 2 (nominal parameter count, descending).

		Llama-3.3-70B					Magistral-Sml (24B)					Phi-4 (14B)					Qwen3-VL-8B					Qwen2.5-7B					Phi-3.5-mini (3.8B)				
ISO	**Region**	MIS_v	MIS_s	$\%\Delta_{\text{MIS}}$	JSD_s	r_s	MIS_v	MIS_s	$\%\Delta_{\text{MIS}}$	JSD_s	r_s	MIS_v	MIS_s	$\%\Delta_{\text{MIS}}$	JSD_s	r_s	MIS_v	MIS_s	$\%\Delta_{\text{MIS}}$	JSD_s	r_s	MIS_v	MIS_s	$\%\Delta_{\text{MIS}}$	JSD_s	r_s	MIS_v	MIS_s	$\%\Delta_{\text{MIS}}$	JSD_s	r_s
ARG	Americas	1.015	.891	+12.2	.178	−0.456	.364	.342	+6.0	.046	.601	.463	.389	+16.1	.046	.412	.520	.339	+35.0	.083	.508	.469	.377	+19.7	.064	.196	.770	.645	+16.1	.159	−0.498
BRA	Americas	.521	.458	+12.2	.119	.222	.411	.319	+22.5	.051	.333	.274	.299	-9.3	.061	.471	.529	.375	+29.1	.101	.212	.580	.411	+29.1	.081	−0.205	.531	.649	-22.1	.173	−0.743
COL	Americas	1.041	.931	+10.5	.189	−0.594	.413	.369	+10.7	.055	.386	.487	.438	+10.1	.057	.093	.559	.390	+30.1	.093	.161	.508	.417	+17.8	.070	−0.096	.788	.678	+14.0	.165	−0.602
MEX	Americas	1.037	.909	+12.3	.179	−0.385	.349	.333	+4.6	.040	.707	.485	.420	+13.4	.044	.512	.517	.350	+32.3	.081	.529	.467	.388	+17.0	.062	.252	.782	.662	+15.4	.159	−0.494
USA	Americas	.873	.578	+33.8	.121	.266	.448	.374	+16.5	.042	.731	.497	.243	+51.1	.058	.723	.677	.487	+28.0	.110	.422	.483	.383	+20.8	.075	.709	.645	.579	+10.2	.110	−0.380
DEU	Europe	.716	.643	+10.1	.189	.101	.265	.344	-30.3	.045	.724	.302	.255	+15.6	.059	.779	.318	.306	+3.7	.081	.568	.308	.415	-34.6	.069	.144	.569	.555	+2.5	.146	−0.403
GBR	Europe	.873	.598	+31.4	.141	.125	.442	.384	+13.2	.048	.674	.503	.319	+36.6	.082	.594	.677	.524	+22.6	.128	.230	.483	.399	+17.4	.082	.620	.654	.588	+10.1	.114	−0.392
ROU	Europe	.851	.593	+30.3	.147	.197	.446	.363	+18.7	.043	.708	.504	.385	+23.5	.058	.570	.659	.535	+18.8	.123	.162	.442	.330	+25.2	.051	.785	.671	.577	+14.0	.117	−0.352
SRB	Europe	.862	.605	+29.9	.150	.157	.463	.371	+19.8	.045	.670	.500	.385	+22.9	.057	.564	.676	.567	+16.2	.127	.075	.453	.303	+33.1	.051	.773	.677	.602	+11.1	.124	−0.463
CHN	E. Asia	.901	.748	+16.9	.183	−0.025	.367	.293	+20.2	.050	.711	.407	.214	+47.4	.054	.781	.569	.415	+27.0	.121	.269	.437	.356	+18.6	.084	.405	.738	.559	+24.3	.162	−0.041
JPN	E. Asia	.521	.471	+9.5	.107	.249	.341	.265	+22.3	.054	.709	.395	.195	+50.6	.051	.671	.407	.316	+22.4	.090	.453	.296	.356	-20.1	.047	.535	.454	.455	-0.3	.063	−0.233
IDN	SE. Asia	.688	.557	+19.0	.092	.071	.283	.344	-21.5	.059	.511	.459	.274	+40.3	.059	.595	.408	.402	+1.5	.091	.318	.480	.402	+16.1	.058	.251	.492	.484	+1.6	.120	−0.769
MMR	SE. Asia	.872	.652	+25.2	.185	−0.031	.496	.365	+26.5	.062	.457	.482	.357	+25.9	.057	.570	.695	.563	+18.9	.139	−0.066	.461	.302	+34.3	.062	.590	.685	.590	+13.8	.135	−0.623
MYS	SE. Asia	.841	.589	+29.9	.155	.003	.443	.313	+29.4	.043	.683	.459	.328	+28.5	.053	.614	.656	.504	+23.1	.128	.097	.445	.274	+38.5	.050	.763	.631	.535	+15.1	.118	−0.426
THA	SE. Asia	.815	.577	+29.2	.152	.125	.406	.295	+27.3	.041	.738	.456	.313	+31.4	.047	.685	.619	.480	+22.5	.112	.231	.414	.221	+46.6	.046	.841	.619	.520	+16.1	.112	−0.283
VNM	SE. Asia	.845	.740	+12.4	.142	−0.045	.324	.347	-7.2	.050	.589	.406	.336	+17.3	.064	.596	.445	.420	+5.5	.087	.339	.501	.468	+6.6	.060	−0.018	.481	.507	-5.4	.067	−0.143
BGD	S. Asia	.864	.628	+27.3	.162	.029	.450	.361	+19.6	.047	.625	.489	.368	+24.7	.048	.607	.674	.587	+12.9	.133	−0.116	.461	.284	+38.3	.055	.695	.666	.565	+15.3	.115	−0.426
KGZ	C. Asia	.849	.585	+31.1	.149	.176	.460	.353	+23.2	.048	.633	.499	.393	+21.1	.058	.476	.661	.574	+13.1	.133	−0.017	.443	.249	+43.8	.039	.829	.660	.572	+13.3	.119	−0.405
IRN	W. Asia	1.039	.852	+17.9	.159	−0.294	.427	.401	+6.1	.055	.637	.397	.530	-33.5	.086	.055	.507	.506	+0.2	.090	.057	.373	.467	-25.1	.067	.252	.511	.445	+12.9	.056	.766
ETH	Africa	.967	.747	+22.7	.172	−0.077	.462	.472	-2.0	.054	.655	.615	.485	+21.2	.041	.808	.724	.684	+5.4	.141	−0.095	.558	.438	+21.6	.059	.676	.734	.661	+10.0	.114	−0.027
Mean		**.849**	**.668**	**+21.2**	**.154**	**−0.009**	**.403**	**.350**	**+11.3**	**.049**	**.624**	**.454**	**.346**	**+22.7**	**.057**	**.559**	**.575**	**.466**	**+18.4**	**.110**	**.217**	**.453**	**.362**	**+18.2**	**.062**	**.450**	**.638**	**.571**	**+9.4**	**.122**	**−0.347**

Legend. MIS_v/MIS_s: vanilla vs. SWA-DPBR ℓ_2 misalignment; $\%\Delta_{\text{MIS}}$: relative improvement in MIS; JSD_s and r_s: Jensen–Shannon distance and Pearson correlation for SWA-DPBR only. **Region groups:** Americas (ARG, BRA, COL, MEX, USA); Europe (DEU, GBR, ROU, SRB); E. Asia (CHN, JPN); SE. Asia (IDN, MMR, MYS, THA, VNM); S. Asia (BGD); C. Asia (KGZ); W. Asia (IRN); Africa (ETH).

single country even when the aggregate row is positive. That heterogeneity is an *insight* about data and reliability weighting, not a failure of the cultural objective—it is exactly the regime where the dual-pass reliability gate should matter most.

5.3 Robustness Suite at a Glance

Finally, Table 4 strengthens confidence that the headline gains are not an artifact of one temperature knob or one preprocessing script: on a fifteen-country Llama/Qwen panel under single-pass SWA-PTIS, sweeps over T_{dec}, uniform and per-category T_{cat}, and dataset variants produce tight JSD bands, the Prospect-Theory kernel beats deterministic shifts, and the pipeline survives reasonable preprocessing. These checks do not replace dedicated dual-pass seed or reliability-scale studies, but they make the "default" configuration look like a stable engineering choice rather than a lucky draw.

Table 4: Robustness suite. Each test independently probes one assumption behind SWA-DPBR; the bootstrap confidence interval on JSD is ± 0.004.

Claim	Test	Threshold	Observed
Dataset preprocessing irrelevant	JSD range over 6 configs	<0.010	**0.0034**
T_{dec} insensitivity	JSD span (Sweep A)	<0.010	**0.0074**
Uniform T_{cat} insensitivity	JSD span (Sweep B)	<0.010	**0.0073**
Per-category T_{cat} insensitivity	JSD span (Sweep C)	<0.010	**0.0077**
PT-IS $>$ best deterministic shift	ΔJSD(CS-Clamp→PT–IS)	>0	**0.003**
WVS weighting alone insufficient	ARGS-WVS $\succ$ ARGS-Unif	$<8/15$	**3/15**
PT–IS kernel $\equiv$ ARGS-PT	ARGS-PT best/tied	$\geq 10/15$	**13/15**
Qwen2.5-32B failure: logit concentration	H vs. H_{ref}=2.5	$H<1$	**0.521** nats
Cross-lingual pipeline generalises	end-to-end runs	8/8	**8/8**

6 What Makes It Work: Ablation

We ablate four design choices on Qwen2.5-72B, USA (342 scenarios; Table 5).[1]

Row 1 shows that dropping PT-IS and keeping only persona consensus hurts JSD and Pearson r materially. Row 2 matches the full system within numerical noise on this slice (Appendix I). Row 3

[1]The ablation uses the full 342-scenario USA pool (JSD $= 0.0425$ for the full configuration). Subsampling scenarios changes absolute JSD but preserves the relative ordering across rows.

Table 5: Ablation on Qwen2.5-72B, USA (342 scenarios), **SWA-DPBR** with one K-sample importance-sampling batch per scenario (dual-pass reliability disabled for isolation). Full configuration: JSD = 0.0425, r = 0.724.

#	Configuration	JSD ↓	ΔJSD	r ↑	Δr	MAE
	Full SWA-DPBR	**0.0425**	–	**0.724**	–	**9.77**
1	No-IS (consensus only)	0.0493	+.007	0.650	−.074	11.02
2	Always-on PT-IS	0.0422	−.000	0.729	+.005	9.74
3	No debiasing	0.0476	+.005	0.743	+.019	9.67
4	Without persona	0.0610	+.019	0.441	−.283	11.09

shows that **positional debiasing** still matters (+12.0% JSD in this table; Species preference inflates by 7.7pp without it). Row 4 removes the WVS-grounded persona panel (base prompt only): JSD rises sharply and Pearson r collapses, confirming that survey personas are load-bearing, not decorative. The IS *flip rate* is only 6.0%, so the framework *modulates confidence* rather than reversing decisions (Appendix J).

7 Discussion

Three takeaways tie the tables together.

(i) Disagreement as signal. SWA-DPBR is a lightweight intervention: culturally grounded personas supply structured disagreement; Prospect-Theory importance sampling turns that structure into a bounded logit nudge; dual-pass reliability shrinks the nudge when the two Monte Carlo estimates disagree. When personas largely agree, corrections stay small because consensus and IS weights line up; when they disagree, the IS stage has room to move. On well-calibrated backbones the alignment gains are large without any weight update.

(ii) Heterogeneity tracks models, not geography. Per-country grids (§5.2) show broad gains across regions; the residual trouble spots line up with sparse persona coverage or collapsed logits, not with a single "hard continent." What correlates with weaker behaviour is *architecture and logit shape*—for example Qwen2.5-32B degrades where decision entropy collapses, matching the diagnosis in Table 4 and Appendix K.

(iii) The Prospect-Theory kernel is load-bearing. Table 5 shows that dropping PT–IS while keeping persona consensus hurts JSD and r; personas alone without the kernel are not enough. The robustness suite (Table 4) reinforces that WVS-linear kernels and plain deterministic shifts underperform the same cooperative utility we encode in Eq. 7.

Dimension hotspots (internal diagnostics). Across logged per-dimension tables, the largest human–model AMCE gaps most often appear on **Species_Humans** and **Utilitarianism_More**, with **Age_Young** frequently among the worst dimensions—consistent with models hugging $\approx 50\%$ sparing rates where humans are extreme. SWA-DPBR improves the aggregate six-dimensional MIS but does not erase those hard cases; they remain the natural stress test for any cross-cultural decoder.

When correction hurts. Internal five-country checks also show a compatibility pattern the macro table understates: if vanilla MIS is already very low for a given (model, country) pair, adding hierarchical or IS-based mass on top can *increase* misalignment (the same pipeline may go $0/5$ wins vs. vanilla even when it succeeds elsewhere). This is not a geographic effect; it is a reminder that inference-time steering requires headroom in the base logits.

Deployment cost is dominated by the batched multi-agent forward pass; the IS loop reuses cached scalars and adds negligible latency (Appendix J).

Limitations. The caveats that most directly bound how far one should generalise from this paper are threefold. First, core hyperparameters (σ, λ_{coop}, η, T_{dec}, T_{cat}, and reliability scale s) were selected on 200 GPT-4-generated dilemmas with author proxy labels and transfer-checked on 50 MultiTP scenarios; the temperature and dataset sweeps in Appendices E–F are our main evidence that these values carry over. The Prospect-Theory parameters (α, β, κ) follow classical monetary risk

literature [Kahneman and Tversky, 1979]; we use them as a smooth gain/loss shaping on logit-gap coordinates—not as a normative claim that moral preference should satisfy the same axioms as monetary gambles. Second, we optimise against country-level AMCE vectors from Moral Machine / MultiTP crowdsourcing [Awad et al., 2018, Jin et al., 2025]; we do not run a human study asking whether SWA-DPBR outputs feel culturally appropriate in context [Atari et al., 2023], so the reader should treat our numbers as alignment to a survey statistic, not a proof of perceived legitimacy. Third, all headline metrics aggregate six moral dimensions; we do not yet report which dimensions improve or worsen. Heterogeneous T_{cat} raises an open question about uneven correction across dimensions—systematic per-dimension analysis is our largest analytical gap. SWA-DPBR also requires decision-token *logits*; hosted APIs that expose only text cannot run the pipeline without a proxy decoder, which limits strict black-box deployment. Additional limitations (single seed, noisy Pearson r, indirect WVS-to-trolley mapping, geographic and temporal coverage) are in Appendix L.

Broader impact and safeguards. SWA-DPBR sits in a tension familiar from pluralistic alignment work [Sorensen et al., 2024]: it can make models more responsive to documented cross-national preference heterogeneity without retraining, yet it can also bake in harmful majorities or survey stereotypes if deployed naively. We suggest four safeguards: governance so SWA-DPBR does not override safety-critical guardrails and domain experts review target cultures; periodic refresh of Wave 7-grounded personas as newer WVS data appear; audits of per-dimension AMCE shifts to catch entrenched bias; and optional clipping or penalty terms on sensitive dimensions inside U_{total} to turn post-hoc review into decoding-time constraints. Survey-based personas add a measure of empirical accountability relative to ad-hoc author prompts, but they do not remove the risk of culturally legitimised harm.

8 Conclusion

Inference-time cultural steering works when the frozen model's decision surface cooperates: SWA-DPBR shows that survey-grounded persona *disagreement* can be turned into a reliable logit correction via dual-pass Prospect-Theory importance sampling with bootstrap damping (Eqs. 10–11) and classical gain/loss shaping [Kahneman and Tversky, 1979]. The limits are not geographic but mechanical-collapsed logits break IS-so the next step is richer proposal adaptation and broader evaluation (per-dimension outcomes, non-binary actions, updated survey waves, and decoding-time safeguards) alongside pluralistic-alignment practice [Sorensen et al., 2024].

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A AMCE Estimation Details

Binary dimensions. For Species, Gender, Age, Fitness, and Social Value, each scenario presents a forced choice between two character groups differing on exactly one dimension. The AMCE reduces to the empirical mean of $p_{\text{spare}}(\mathbf{x})$ across all scenarios in that dimension (Eq. 1), consistent with MultiTP conventions [Jin et al., 2025].

Utilitarianism dimension. Utilitarianism scenarios vary the *number* of lives on each side rather than character identity. Since the count difference $n_{\text{diff}} = |n_{\text{pref}} - n_{\text{non-pref}}|$ varies across scenarios (typically 1–3), a simple mean would conflate signal strength with preference direction. We therefore fit an OLS regression $p_{\text{spare}} = a + b \cdot n_{\text{diff}}$ and evaluate at the mean observed n_{diff} in the data. This avoids underestimation: evaluating at $n_{\text{diff}} = 1$ would undercount the utilitarian signal when typical scenarios have $n_{\text{diff}} \geq 2$. Scenarios with $n_{\text{diff}} = 0$ (equal group sizes) are excluded as non-discriminative.

Human AMCE scale. The MultiTP ground-truth AMCEs in `country_specific_ACME.csv` are on a $[-1, 1]$ scale. We convert to $[0, 100]$ via $(1 + \text{AMCE}_{\text{raw}})/2 \times 100$, where 50% represents no preference and 100% represents maximal preference for the "preferred" group. Model AMCEs are computed on the same scale.

Consistency. Awad et al. [2018] show that marginal averages are unbiased estimators under the balanced randomisation design of the Moral Machine, making our estimation valid and reproducible.

B Persona Construction Details

B.1 Example Personas

The young-adult persona for the **United States**:

> *You are a young adult (18–35) from the United States of America. Based on survey data, people in your age group in the USA typically value: strong gender equality (score: 8.2/10), low religious centrality in moral decisions (score: 3.1/10), high personal autonomy (score: 7.8/10), high meritocratic orientation (score: 7.5/10), moderate family importance (score: 6.4/10), and moderate interpersonal trust (score: 5.3/10). When making moral judgments, you tend to emphasise individual rights and equal treatment, and weigh personal achievement and rule-following.*

The older-adult persona for **Vietnam**:

> *You are an older adult (55+) from Vietnam. Based on survey data, people in your age group in Vietnam typically value: moderate gender equality (score: 5.1/10), high religious and moral traditionalism (score: 7.2/10), strong family importance (score: 9.4/10), moderate personal autonomy (score: 4.8/10), and low interpersonal trust toward strangers (score: 3.2/10). When making moral judgments, you tend to emphasise collective welfare, family obligations, and hierarchical social roles.*

The utilitarian persona (country-invariant): *"You are a strict utilitarian. Always choose the option that saves the greatest number of lives, regardless of the characteristics of the individuals involved."*

B.2 WVS Data Processing Pipeline

We process WVS Wave 7 individual-level microdata as follows:

1. **Variable extraction:** For each respondent, we extract 8 value dimension scores from specific WVS items: gender egalitarianism (mean of Q58P, Q59P, Q60P), religious importance (Q6P), interpersonal trust (Q43P), moral permissiveness (mean of Q50, Q52P, Q54P), work centrality (Q5P), family importance (Q1P), personal autonomy (Q39P), meritocratic orientation (Q40P). All items use inverted scales where higher = more progressive/positive.
2. **Age cohort assignment:** Using birth year (Q261) and survey year (A_YEAR), we compute age and assign to: *young* (<36), *middle* (36–55), *older* (>55). Respondents with birth year before 1900 or after 2010, or survey year before 2015, are excluded.

3. **Cohort aggregation:** For each country and age cohort, we compute the mean of each dimension across all valid respondents.
4. **Natural-language conversion:** Dimension means are converted to descriptors using fixed thresholds, listed in full in Table 6. Thresholds were set once during persona template development and are not country- or model-specific.
5. **Fallback:** For countries with insufficient WVS coverage (e.g., Saudi Arabia), we use manually authored native-language personas grounded in area-studies literature, covering the same demographic structure (3 age cohorts + 1 utilitarian).

Table 6: Complete WVS-dimension-to-descriptor threshold table used by the persona generator. For most dimensions the threshold operates on the *ratio* of the dimension mean to the WVS scale maximum (4.0); the exception is moral permissiveness, which operates on the raw 1–10 mean because its underlying items use a different scale. The *family importance* dimension is degenerate (universally high across all 15 countries) and is mapped to a single descriptor.

Dimension (input)	Cut-off	Descriptor
Religiosity (ratio)	> 0.85	deeply religious
	> 0.70	moderately religious
	> 0.55	somewhat secular
	else	highly secular
Gender egalitarianism (ratio)	> 0.85	strongly gender-egalitarian
	> 0.75	moderately gender-egalitarian
	> 0.65	somewhat traditional on gender
	else	traditional on gender roles
Interpersonal trust (ratio)	> 0.55	high interpersonal trust
	> 0.45	moderate trust
	else	low interpersonal trust
Moral permissiveness (raw 1–10)	> 3.5	morally permissive
	> 3.0	moderately permissive
	> 2.5	morally conservative
	else	morally strict
Personal autonomy (ratio)	> 0.90	strongly values personal autonomy
	> 0.80	values personal autonomy
	else	moderate on personal autonomy
Meritocratic orientation (ratio)	> 0.95	strongly meritocratic
	> 0.85	meritocratic
	else	egalitarian on income
Work centrality (ratio)	> 0.90	work is central to identity
	> 0.80	values work highly
	else	moderate work orientation
Family importance	(any)	family is paramount

B.3 WVS-to-Trolley Dimension Linkage

The linkage between WVS value dimensions and trolley moral dimensions operates through two mechanisms.

Direct thematic overlap: WVS *gender egalitarianism* maps onto the Gender dimension (societies scoring higher show weaker female-sparing preferences [Awad et al., 2018]); *religious importance* correlates with Species (human exceptionalism); *meritocratic orientation* relates to Social Value and Fitness.

Indirect modulation: Dimensions like *personal autonomy*, *interpersonal trust*, and *family importance* modulate the moral *frame*; e.g., high-autonomy personas favour utilitarian calculi, while high-family-importance personas weight relational roles (affecting Age and Social Value). Crucially, SWA-DPBR does not require exact causal mapping: the ensemble generates diversity in logit-space gaps, and the

IS stage selects the correction balancing collective utility. A rigorous per-dimension causal analysis (e.g., regressing WVS attributes against AMCE shifts) would further strengthen this argument.

C PT–IS, Sampling, and Dual-Pass Reliability

C.1 Utility shape and the omitted explicit KL-on-ϵ term

The scalar decision gap δ acts as a one-dimensional state and the perturbations ϵ_k as control inputs at a single horizon step. Under the path-integral / free-energy view [Williams et al., 2018], softmax weights $w_k \propto \exp(U_{\text{total}}/\eta)$ can be read as targeting a Boltzmann distribution over candidate perturbations. **Our released code does not add** a separate quadratic penalty $\propto \epsilon_k^2/(2\sigma^2)$ to U_{total} in Eq. 7: the released controller matches it exactly, and the Gaussian proposal $\epsilon_k \sim \mathcal{N}(0, \sigma^2)$ already limits how far $\tilde{\delta}_k$ can move from $\bar{\delta}$. An optional explicit KL-on-control term remains a straightforward extension if future work needs sharper tail control on $|\epsilon_k|$.

C.2 Comparison with Simpler Alternatives

Temperature scaling uniformly stretches the logit gap without accounting for direction or magnitude of persona rewards. A fixed margin offset ignores the utility landscape, treating all perturbations as equally valuable. PT-weighted IS performs an *adaptive, scenario-specific* correction via nonlinear importance weighting. Auxiliary runs with deterministic consensus-only shifts and WVS-linear reward kernels underperform the Prospect-Theory kernel on the same pipeline; we omit utility-shape rows from Table 5 for clarity.

C.3 Why split K into two passes?

Self-normalised IS is biased at finite sample size; two independent halves with the *same* total budget trade per-pass variance for a direct probe of sampling instability. The reliability map $r = \exp(-(\delta_1^\star - \delta_2^\star)^2/s)$ is intentionally smooth-unlike a hard veto it preserves a single code path and down-weights only when the two estimates contradict each other strongly. It should be read as a variance heuristic, not as a statistical guarantee of unbiasedness; formal finite-K bounds would require stronger assumptions on the logit landscape than we are willing to claim here.

C.4 Positional Debiasing: Implementation Detail

The two-pass debiasing of §3.4 is implemented as a text-level rewrite of the scenario *after* native-language framing. To avoid the trivial bug of double-substitution (replacing LEFT $\rightarrow$ RIGHT immediately followed by RIGHT $\rightarrow$ LEFT would leave every label as LEFT), we use a placeholder-based three-step swap, applied independently for the lane-label pair and the group-label pair:

1. Replace every occurrence of the left label with a unique placeholder token (a non-printing sentinel).
2. Replace every occurrence of the right label with the left label.
3. Replace the placeholder with the right label.

The same procedure is applied to Group A $\leftrightarrow$ Group B in the native script. Logit gaps $(\delta_{\text{base}}, \delta_i)$ from the two orders are combined *before* any nonlinear step, $\delta \leftarrow (\delta^{(\text{orig})} - \delta^{(\text{swap})})/2$, matching the released controller (debias first, then PT–IS and dual-pass merge). This guarantees cancellation of additive positional bias in logit space; the final p_{spare} applies $\sigma(\cdot)$ only after that correction.

D Hyperparameter Validation

Implementation. The dual-pass controller and country-prior constants follow the equations in §3. Optional environment overrides can change K_{half}, reliability scale s, and whether the ESS-anchor blend is active; such flags must be applied before controller initialisation when running ablations.

Non-PT hyperparameters (σ, λ_{coop}, η, T_{dec}, T_{cat}) were validated on a held-out synthetic set of 200 binary moral dilemmas generated by GPT-4, covering the same six dimensions as MultiTP with

Table 7: SWA-DPBR hyperparameters. Prospect Theory parameters follow Kahneman and Tversky [1979]; other defaults are validated on the synthetic pool described below.

Parameter	Symbol	Value
Persona agents	N	4
IS samples / pass	K_{half}	64 (total $2K_{\text{half}} = 128$)
Reliability scale	s	0.04
Perturbation std. dev.	σ	0.3
Cooperation weight	λ_{coop}	0.7
IS softmax temp.	η	0.5
ESS guard threshold	ρ_{eff}	0.1
PT curvature	$\alpha{=}\beta$	0.88
PT loss aversion	κ	2.25
Decision temp.	T_{dec}	0.5
T_{cat} (Species / Gender / other)	-	4.0 / 3.5 / 1.5
Default logit divisor	T_{logit}	3.0
Country prior	$N_{\text{warm}}, \tau, \beta_{\text{ema}}$	50, 100, 0.1
ESS anchor blend	-	on (off: $\bar{\delta} + \delta^{\star}_{\text{IS}}$ only)

novel character descriptions. Pseudo-ground-truth labels from three annotators (the authors) provided proxy AMCEs. Grid search minimised mean JSD on Qwen2.5-72B. Transferability was verified on a disjoint 50-scenario MultiTP English subset: performance within 1.2% relative JSD of oracle-tuned configuration. T_{cat} values reflect empirical logit magnitude distributions: Species and Gender produce systematically larger gaps, requiring higher temperatures. Because this tuning pool is synthetic and author-annotated, residual bias is possible; we therefore report broad robustness sweeps (Appendix E, Appendix F) to show conclusions are stable beyond one tuned point.

E Temperature Sensitivity

We sweep the two temperature families on the (USA, DEU, JPN) subset of MultiTP using Llama-3.1-70B. Sweep A varies $T_{\text{dec}} \in \{0.10, 0.25, 0.50, 0.75, 1.0, 2.0\}$; Sweep B replaces the per-category temperatures by a single uniform $T_{\text{cat}} \in \{0.5, 1.0, 1.5, 2.0, 3.0, 4.0, 6.0\}$; Sweep C varies the "Others" bucket $T_{\text{cat}} \in \{0.75, 1.0, 1.5, 2.0, 3.0, 4.0\}$ while holding $T_{\text{cat}}[\text{Species}]{=}4.0$, $T_{\text{cat}}[\text{Gender}]{=}3.5$. All three sweeps yield JSD spans below 0.010 ($0.0074,\ 0.0073,\ 0.0077$ respectively), and in every sweep the SWA-DPBR default (†) is strictly *more conservative* than the empirically best value ($\Delta\text{JSD} \in \{-0.0061, -0.0021, -0.0037\}$). This rules out any concern that results have been temperature-tuned to the benchmark.

The monotonic JSD reduction with larger T_{dec} reflects the role of T_{dec} in undoing RLHF logit compression: higher values widen the effective token-probability landscape so that IS perturbations exert larger directional influence. However, r peaks near $T_{\text{dec}}{=}1.0$ and declines at 2.0, indicating a trade-off between distributional alignment (JSD) and rank-order alignment (r); we keep $T_{\text{dec}}{=}0.5$ as a deliberately conservative operating point.

F Dataset Preprocessing and Sensitivity

Pipeline. Each MultiTP CSV is processed in five steps: (i) deduplication keeps only the canonical paraphrase (`which_paraphrase = 0`); (ii) the Utilitarianism *quality filter* drops scenarios where the two groups have identical size *and* all characters belong to the *quality-attribute* role set {PREGNANT, WOMAN, LARGEWOMAN}, which would otherwise inject a non-numerosity signal into a dimension whose AMCE is by construction the count-difference effect; (iii) per-category capping retains at most 80 scenarios per dimension; (iv) under-represented categories are oversampled with replacement to a minimum of 36 scenarios; and (v) the resulting pool is shuffled with a fixed seed (42).

Reproducible left/right assignment. When the MultiTP `paraphrase_choice` field does not unambiguously determine which group is rendered on the left vs. right, we deterministically fall back to a SHA-256 hash of the tuple (sub_1, sub_2, g_1, g_2) modulo 2, so the same scenario receives the same

Table 8: Temperature sensitivity. Defaults (†) are strictly worse than the best setting in every sweep, providing a conservative lower bound.

Sweep	Setting	JSD ↓	Pearson r ↑	ΔJSD
A: T_{dec}	0.10	0.0502	0.662	+0.0013
	0.25	0.0491	0.658	+0.0002
	0.50†	0.0489	0.630	-
	0.75	0.0476	0.636	−0.0013
	1.00	0.0466	0.641	−0.0023
	2.00 (best)	0.0428	0.627	−0.0061
	JSD span		**0.0074**	
B: Uniform T_{cat}	0.5	0.0521	0.642	+0.0052
	1.0	0.0492	0.673	+0.0023
	1.5†	0.0469	0.700	-
	2.0	0.0461	0.707	−0.0008
	3.0	0.0454	0.707	−0.0015
	4.0	0.0451	0.701	−0.0018
	6.0 (best)	0.0448	0.684	−0.0021
	JSD span		**0.0073**	
C: Others T_{cat}	0.75	0.0526	0.566	+0.0039
	1.00	0.0512	0.589	+0.0025
	1.50†	0.0487	0.634	-
	2.00	0.0475	0.660	−0.0012
	3.00	0.0458	0.693	−0.0029
	4.00 (best)	0.0450	0.706	−0.0037
	JSD span		**0.0077**	

ordering across runs and machines. The fallback rate is logged per country; for all 15 countries it stays below 5%.

Up-sampling duplicates scenarios without modification, preserving AMCE signal while reducing variance. Category capping prevents dimension dominance. Side balancing enables effective debiasing. To evaluate whether these choices alter the effective evaluation distribution, we compare *six* preprocessing configurations on 10 representative countries (USA, DEU, CHN, JPN, BRA, VNM, GBR, KOR, RUS, NGA) using Llama-3.1-70B.

Table 9: Per-country dataset sensitivity. JSD range = 0.0034 (aggregate) and < 0.010 in 9/10 countries; the only exception is RUS at 0.013, driven by augmentation sensitivity in the rare Species/Utilitarianism categories. D0 is retained as default.

Country	D0-Default	D1-NoAug	D2-Cap40	D3-Cap120	D4-NoFlip	D5-Strict
USA	0.0655	0.0674	0.0625	0.0644	0.0643	0.0684
DEU	0.0465	0.0515	0.0492	0.0467	0.0444	0.0484
CHN	0.0393	0.0363	0.0385	0.0383	0.0409	0.0398
JPN	0.0304	0.0291	0.0319	0.0296	0.0273	0.0320
BRA	0.0492	0.0551	0.0479	0.0510	0.0520	0.0480
VNM	0.0592	0.0574	0.0631	0.0585	0.0578	0.0626
GBR	0.0614	0.0667	0.0608	0.0618	0.0597	0.0633
KOR	0.0371	0.0354	0.0378	0.0357	0.0349	0.0382
RUS	0.0383	0.0489	0.0389	0.0374	0.0383	0.0357
NGA	0.0522	0.0568	0.0506	0.0525	0.0514	0.0538
Mean	**0.0479**	0.0505	0.0481	0.0476	0.0471	0.0490
Δ vs D0	-	+0.0025	+0.0002	−0.0003	−0.0008	+0.0011

D1-NoAug shows the largest degradation (+0.0025), justifying synthetic augmentation of the under-represented Species and Utilitarianism categories. D4-NoFlip is marginally lower than D0, indicating that group-side randomisation is conservative (it adds noise) rather than harmful; we retain it because it preserves positional-debiasing integrity. The 0.0034 aggregate range is well below the ± 0.004 bootstrap CI, so reported results are not confounded by preprocessing choices.

G Cross-Lingual Token Elicitation

Native-language prompting system. Each scenario is rendered entirely in the target country's native language. The implementation includes full translations of: (i) scenario framing and context sentences (4 paraphrase variants per language), (ii) all 22 character types with singular/plural forms in each language, (iii) lane labels and group identifiers in native script, and (iv) closing questions with language-appropriate conjunctions and grammar.

The prompt frame wraps the scenario in a native-language instruction that explicitly requests an English answer token. Each frame follows the pattern: "[moral dilemma preamble in native language] \n {scenario} \n [instruction to answer A or B in English]". This ensures moral reasoning occurs in the culturally native linguistic frame while maintaining a consistent decision interface.

Token verification. Token IDs are verified per model at initialisation by encoding the uppercase strings `"A"` and `"B"` with `add_special_tokens=False` and asserting single-token mappings: Qwen2.5: A=22397, B=11572; Llama-3.1: A=31266, B=9268. Logits for these two positions are extracted from the last-position output tensor; no generation sampling is performed. The model uses its built-in `chat_template` for correct role formatting across model families.

H Baseline Implementation Details

H.1 Activation Steering

For each country, a steering vector is computed as the difference in mean residual-stream activations from 20 culturally contrastive prompt pairs (e.g., "Answer as someone from [country] who values [WVS-high trait]" vs. "Answer as someone with the opposite values"). Vectors are extracted from layer 32 (transformer midpoint) of Llama-3.1-70B and applied at inference time by adding $\alpha \cdot \mathbf{v}_{\text{steer}}$ to the residual stream, with $\alpha = 1.5$ selected via the same synthetic validation set used for SWA-DPBR.

Sensitivity analysis and fairness considerations. Activation steering performance is known to be sensitive to (i) the extraction layer, (ii) the scaling coefficient α, and (iii) the construction of contrastive pairs [Arditi et al., 2025]. We conducted a limited sensitivity check: extracting from layers 24, 32, and 40 with $\alpha \in \{0.5, 1.0, 1.5, 2.0, 3.0\}$, we found that layer 32 with $\alpha = 1.5$ yielded the lowest mean JSD (0.0877); other configurations ranged from 0.089 to 0.112. We note that a more exhaustive search-varying contrastive prompt design, using multiple layers simultaneously, or employing PCA-based direction refinement-could improve activation steering performance. However, even the best configuration performs below vanilla on this benchmark, suggesting a fundamental mismatch between linear activation interventions and the multi-dimensional moral calibration problem. The key structural advantage of SWA-DPBR is that it operates *per-scenario* with adaptive correction magnitude, whereas activation steering applies a fixed direction uniformly across all scenarios.

H.2 ARGS and controlled decoding (relation to our work)

ARGS [Khanov et al., 2024] and controlled decoding [Mudgal et al., 2024] typically assume trained reward models; a cross-cultural deployment would require a separate reward per target country. On binary forced-choice MultiTP, one can instead define rewards as closed-form functionals of persona logit gaps; a Prospect-Theory kernel aligned with Eq. 7 is then the natural analogue of importance-weighted PT–IS. We do not run the full ARGS decoding stack in our experiments; Table 4 and §6 isolate the kernel comparison on the shared SWA-DPBR pipeline.

I Detailed Ablation Analysis

Without persona: Row 4 removes all WVS-conditioned persona prompts (base-only moral elicitation on the same scenarios). The large JSD lift and Pearson r drop in Table 5 match the expected loss of within-country diversity and numerosity anchoring from the persona ensemble.

Debiasing: Mean positional bias is 0.193 logit units; Species preference inflates from 85.8% to 93.5% without it (human: 79.2%).

J Trigger Mechanism and Latency

SWA-DPBR runs dual-pass IS on every scenario; inter-persona variance is logged but does not gate sampling. The *flip rate* (IS reversals vs. consensus) is 6.0% in the USA ablation (Table 5), indicating confidence modulation rather than wholesale decision reversal.

Mean latency: 730ms (SWA-DPBR) vs. 160ms (vanilla) on Qwen2.5-72B (H100). Overhead from batched multi-agent passes and debiasing; IS operates on cached scalars and contributes negligibly.

K Entropy-Aware Adaptive σ for Qwen2.5-32B

We study whether an entropy-aware adaptive σ can address the Qwen2.5-32B failure mode (Section 5). At each scenario we measure the Shannon entropy H of the top-50 token distribution at the decision position and set $\sigma = \sigma_{\text{base}} \cdot \text{clip}(H/H_{\text{ref}},\ 0.2,\ 5.0)$ with H_{ref}=2.5 nats and σ_{base}=0.30. We compare against a fixed-σ sweep on 8 countries (USA, DEU, CHN, JPN, BRA, VNM, GBR, KOR).

Table 10: Per-country JSD under fixed-σ on Qwen2.5-32B. Two latent country clusters emerge: USA/JPN/GBR/KOR strongly prefer high σ (USA: $0.0523 \rightarrow 0.0317$ at σ=1.0); DEU/CHN/BRA/VNM strictly prefer low σ. All 8 countries share $H \approx 0.52$ nats, so the entropy signal alone cannot discriminate them.

σ	USA	DEU	CHN	JPN	BRA	VNM	GBR	KOR	Mean
0.05	0.0515	0.0647	0.0533	0.0690	0.0692	0.0710	0.0500	0.0670	0.0620
0.10	0.0519	0.0648	0.0532	0.0684	0.0694	0.0711	0.0499	0.0668	0.0619
0.15	0.0522	0.0649	0.0530	0.0676	0.0698	0.0711	0.0499	0.0665	0.0619
0.20	0.0518	0.0651	0.0527	0.0661	0.0702	0.0713	0.0498	0.0661	0.0616
$0.30^{\dagger}$	0.0523	0.0654	0.0516	0.0644	0.0710	0.0717	0.0491	0.0649	0.0613
0.50	0.0487	0.0661	0.0515	0.0601	0.0732	0.0734	0.0476	0.0604	0.0601
0.80	**0.0427**	0.0672	0.0599	**0.0585**	0.0780	0.0757	**0.0439**	**0.0532**	**0.0599**
1.00	<u>0.0317</u>	0.0681	0.0714	0.0637	0.0833	0.0744	0.0458	<u>0.0522</u>	0.0613
Adaptive ($\bar{\sigma}$=0.086)	0.0516	0.0647	0.0532	0.0688	0.0693	0.0711	0.0500	0.0670	0.0620

The mean entropy is $H = 0.521$ nats, far below H_{ref}, confirming concentrated logits. Adaptive σ self-scales to $\bar{\sigma} = 0.086$ ($r(H, \sigma)$=0.948), validating the diagnostic but collapsing onto the low-σ regime. The best fixed σ=0.80 reaches JSD 0.0599 vs. 0.0613 at default-a marginal absolute gain-while the per-country breakdown exposes the deeper problem: countries cluster into a high-σ group (USA/JPN/GBR/KOR) and a low-σ group (DEU/CHN/BRA/VNM), but *all* countries share $H \approx 0.52$ nats, so a per-scenario entropy signal cannot discriminate them. A complete fix therefore requires richer per-country proposal adaptation-left as future work.

L Extended Limitations

Single seed and stochasticity. All main stochastic runs use one seed per model; each scenario runs two independent IS passes of K_{half}=64 proposals (128 total), matching the released pipeline. That dual-pass design provides an internal reliability signal (Eq. 10) but **not** multi-seed confidence intervals for macro MIS; we therefore report headline gains as point estimates and treat seed-level CIs as a priority for future work.

Pearson r in six dimensions. As noted in Section 5, the AMCE vector is short, so per-country r has large sampling error and small row-wise r deltas are not necessarily statistically separable; primary claims use MIS, with JSD as a secondary diagnostic.

WVS-to-trolley linkage. The mapping from WVS value dimensions to trolley moral dimensions follows the indirect-modulation construction in Appendix B.3, not a fully validated causal chain; we therefore avoid claims of the form "WVS dimension X caused AMCE shift Y."

Inter-persona reward assumption. The reward $r_i = \delta_i - \delta_{\text{base}}$ assumes persona shifts approximate human target directions. This is plausible when the base model represents a WEIRD-biased prior, but if a persona's shift is orthogonal to the human target, the signal may mislead. Per-dimension validation

is needed. Quantifying whether persona-consensus directions correlate with country-level human AMCE vectors independently of the base model—and when corrections diverge from targets—is also open; we do not report that correlation matrix here.

Geographic coverage, preprocessing, and survey vintage. We report 20 countries in the main MIS sweep (Table 2) out of 100+ MultiTP countries; results depend on the preprocessing pipeline in Appendix F. WVS Wave 7 ($\sim$2017–2022) is a fixed historical slice; persona quality is weakest where coverage is sparse (e.g., Saudi Arabia, Brazil), as reflected in per-country tables. No low-resource languages (Amharic, Khmer, Yoruba) appear in the panel; cross-lingual generalisation to such settings remains untested.

Binary format. The logit-difference formulation requires generalisation to softmax for multi-option settings.

Language–culture conflation. One language per country may conflate linguistic and cultural effects.

Model quantisation. 4-bit/Int8 quantisation may introduce logit artefacts affecting IS stability.

WVS temporal staleness. Wave 7 ($\sim$2017–2022) data may not reflect current preferences in rapidly evolving societies.

Qwen2.5-32B failure mode. Concentrated logit distributions ($H{=}0.521$ nats) cause IS perturbations to land off-manifold and yield near-uniform importance weights. Appendix K shows that an entropy-aware σ correctly diagnoses the cause but cannot fix it on its own, because all 8 evaluated countries share the same low entropy and thus cannot be discriminated by a per-scenario entropy signal. A complete remedy requires richer jointly adaptive σ (and related proposal knobs) per country and is left to future work.